AGRIVIVE / KATTA-LOCK

KARNATAKA / KATTA-INSPIRED / INTERLOCKING RUNOFF BARRIER

SERIES 11 / 28

A3 / PRELIMINARY / ALL DIMENSIONS mm



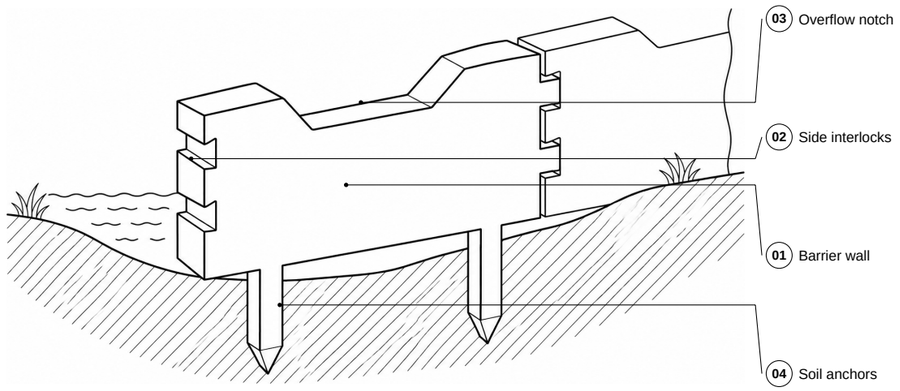
STATE CONTEXT / INDIGENOUS INSPIRATION

Temporary kattas are community water barriers documented in coastal Karnataka and Kerala. KATTA-LOCK proposes a small modular runoff barrier with soil anchors and an open overflow notch; field loads remain unverified.

Practice source: NLSIU Traditional Kattas study | Outline: geoBoundaries / schematic

01 INSTALLED SOIL / WATER SECTION

NOMINAL ENVELOPE / mm: 200 span x 120 barrier H x 30 thick

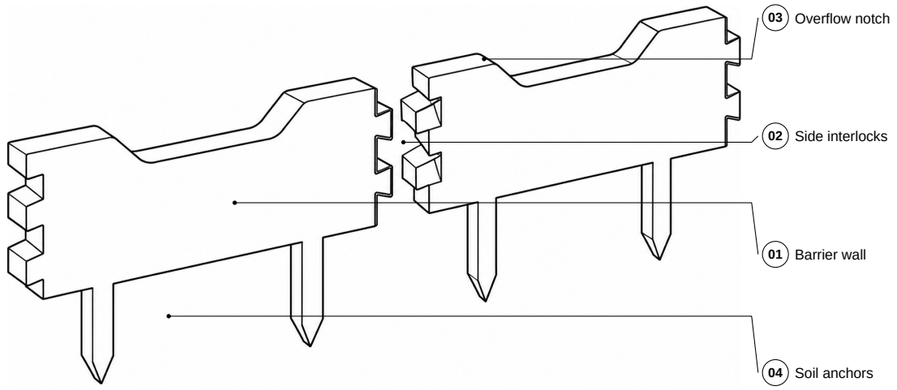


DESIGN / OPERATION NOTES

Interlock adjacent modules along a shallow runoff groove.
Set anchors in suitable soil, keeping the open notch unobstructed.
Excess water passes over the notch; no flow route exists through the solid wall.

02 EXPLODED ASSEMBLY

ASSEMBLED ENVELOPE / mm
200 span x 120 barrier H x 30 thick

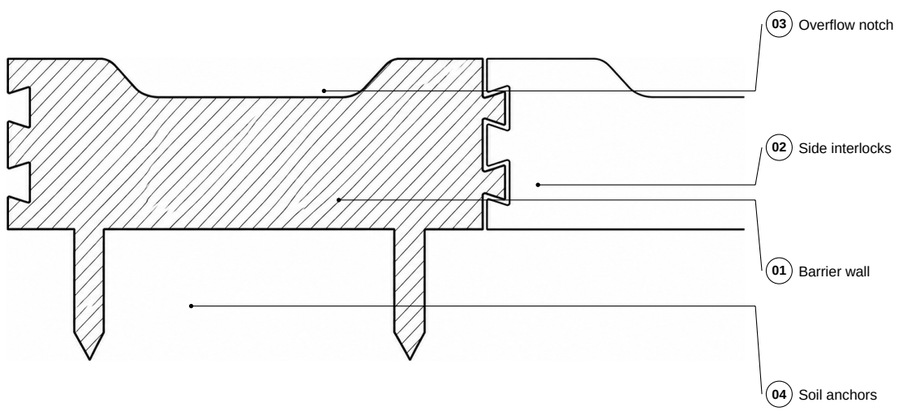


DESIGN / OPERATION NOTES

Components separated for service and assembly.
Connections and mating fits require CAD.

04 INTERNAL SECTION

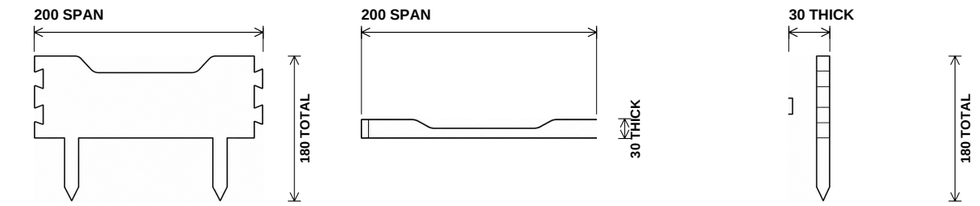
SECTION BASIS / mm
20 side interlock engagement; 60 soil anchor extension



DESIGN / OPERATION NOTES

Set anchors in suitable soil, keeping the open notch unobstructed.
Excess water passes over the notch; no flow route exists through the solid wall.

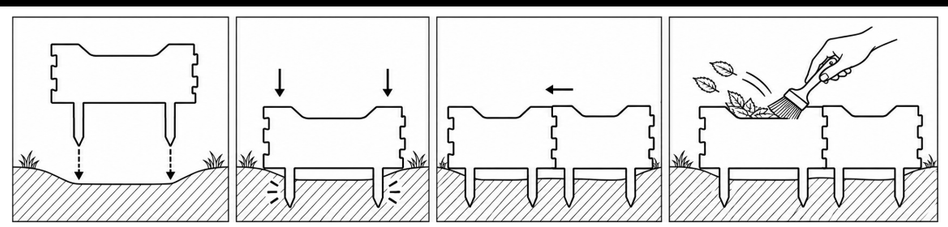
03 ORTHOGRAPHIC STUDIES / mm



FRONT / MODULE **TOP** **SIDE**

Module 200 span x 120 barrier H x 30 thick; two anchors extend 60 below wall. Total H 180.

05 INSTALLATION / OPERATING SEQUENCE



1 ALIGN GROOVE **2 SET ANCHORS** **3 JOIN MODULES** **4 INSPECT NOTCH**

Prototype fit, access and operation require trials.

DESIGN / INTERFACE SCHEDULE

01 NOMINAL GEOMETRY

Module span / barrier H	**200 / 120**
Wall thickness	**30**
Anchor extension	**60**
Interlock engagement	**20**

02 GEOMETRIC CHECK

DESIGN CHECK PENDING

TO BE TESTED

120 barrier H + 60 anchor = 180 total H
Anchorage and runoff loads require soil/site trials

03 PARTS / QTY

01 Barrier wall	**1**
02 Side interlocks	**Integral**
03 Overflow notch	**Integral**
04 Soil anchors	**2 integral**

Wall/anchors: candidate polymer, single-piece module.
Print direction and wet-soil strength unverified.

04 INTERFACES / CHECKS

Check uplift, scour and soil pull-out in site trials.

Specify notch geometry and maximum operating head.

Check interlock clearance and safe bypass route.

PRELIMINARY / VALIDATION PENDING

DESIGN DEVELOPMENT RECORD

01 / GEOMETRY
Nominal body sizes are proposed. Generated views are illustrative; small features and interfaces may vary.

02 / BEFORE FABRICATION
Develop one CAD model, assign wall thicknesses, tolerances and mating fits. Check print orientation and service access.

03 / TRIAL RECORD
Measure dimensions and retained volume or opening area. Record leakage, clogging and soil anchorage under observed conditions.

No discharge, treatment or durability performance is established.

DRAWING / REVISION	TITLE	STATUS	REVISION DATE	FORMAT / SCALE / UNITS	COLLECTION SHEET
AGR-11 / P01	KATTA-LOCK	PRELIMINARY CONCEPT / CAD & TRIALS PENDING	10 NOV 2026	A3 / NTS / mm	11 / 28